



AI Launchpad Program

12-Week Industry-Focused Beginner AI & Python Training Program

This program is specially designed for beginners who are starting their journey in programming, Artificial Intelligence, Machine Learning, Data Analysis, and software development from scratch. The course focuses on practical learning, mini projects, real-world exposure, portfolio building, and confidence development.

Program Information	Details
Program Duration	12 Weeks
Lecture Schedule	Monday – Friday
Lecture Duration	1 Hour Daily
Assignments	Weekend Practice Assignments
Final Outcome	Python + SQL + Machine Learning + Portfolio + Final Project

Mobile Platforms for Students Without Laptop

- Pydroid 3 – Run Python programs directly on Android mobile
- Google Colab – Cloud-based Python notebook accessible through browser
- Replit – Online coding platform for Python projects
- GitHub Mobile – Upload and manage projects from mobile
- SQLite Mobile Apps – Practice SQL queries using mobile
- Canva Mobile – Resume and presentation design



Week 1 – Introduction to Programming & Python Basics

Week Overview

Students will understand the basics of programming, Artificial Intelligence, Python setup, variables, operators, and conditional statements.

Day	Topics Covered
Monday	Introduction to AI, Programming Concepts, Real-Life AI Examples
Tuesday	Python Setup, VS Code/Pydroid, First Python Program
Wednesday	Variables, Data Types, User Input
Thursday	Operators, Arithmetic Logic, Simple Programs
Friday	if-else Conditions, Mini Login System

Weekend Assignment Overview

- Install Python or Pydroid 3 successfully and run 5 sample programs.
- Create a student information program using variables and input.
- Build an even/odd checker and marks grading system.
- Practice at least 10 beginner-level Python problems.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 2 – Loops & Functions

Week Overview

Students will learn logical thinking using loops, functions, and problem-solving techniques.

Day	Topics Covered
Monday	for Loops and Number Patterns
Tuesday	while Loops and Logic Building
Wednesday	Functions and Parameters
Thursday	Nested Conditions and Problem Solving
Friday	Mini Project – Calculator Application

Weekend Assignment Overview

- Create multiplication tables using loops.
- Build a number guessing game.
- Create reusable calculator functions.
- Practice 15 loop and logic-building questions.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 3 – Data Structures

Week Overview

Students will understand how to store and manage data using lists, tuples, sets, dictionaries, and strings.

Day	Topics Covered
Monday	Lists and List Operations
Tuesday	Tuples and Sets
Wednesday	Dictionaries and Student Records
Thursday	Strings and String Manipulation
Friday	Mini Project – Quiz Application

Weekend Assignment Overview

- Create a contact management system using dictionaries.
- Build a shopping list application.
- Practice string manipulation programs.
- Improve the quiz application with scoring logic.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 4 – File Handling & Error Handling

Week Overview

Students will learn how to create real-world applications using files, modules, and exception handling.

Day	Topics Covered
Monday	File Read and Write Operations
Tuesday	Exception Handling
Wednesday	Python Modules and Libraries
Thursday	Project Structuring and Code Organization
Friday	Mini Project – Expense Tracker

Weekend Assignment Overview

- Create a notes manager application.
- Store expense data inside text files.
- Add proper error handling in calculator app.
- Practice importing and using Python libraries.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 5 – SQL Basics

Week Overview

Students will learn how databases work and how SQL is used in real-world applications.

Day	Topics Covered
Monday	Introduction to Databases and SQL
Tuesday	CREATE TABLE and INSERT Queries
Wednesday	SELECT and WHERE Queries
Thursday	UPDATE and DELETE Queries
Friday	Mini Project – Student Database System

Weekend Assignment Overview

- Create database tables for student records.
- Insert, update, and delete records.
- Write at least 20 SQL practice queries.
- Create a small student database system.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 6 – Advanced SQL & Python Integration

Week Overview

Students will connect Python applications with SQL databases and build database-driven systems.

Day	Topics Covered
Monday	ORDER BY and LIMIT
Tuesday	GROUP BY and Aggregate Functions
Wednesday	SQL Joins
Thursday	Python + SQLite Integration
Friday	Mini Project – Library Management System

Weekend Assignment Overview

- Connect Python programs with SQLite database.
- Practice aggregate functions and joins.
- Enhance student database system.
- Build a basic library management application.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 7 – NumPy & Data Analysis

Week Overview

Students will enter the Data Science domain using NumPy arrays and Pandas DataFrames.

Day	Topics Covered
Monday	Introduction to NumPy Arrays
Tuesday	Array Operations
Wednesday	Pandas DataFrames
Thursday	Reading CSV Files
Friday	Mini Project – IPL Dataset Analysis

Weekend Assignment Overview

- Analyze a student marks dataset.
- Practice NumPy array operations.
- Load and filter CSV files using Pandas.
- Create summary reports from datasets.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 8 – Data Cleaning & Visualization

Week Overview

Students will learn how to clean datasets and create charts for analysis and presentation.

Day	Topics Covered
Monday	Cleaning Missing Data
Tuesday	Filtering and Sorting
Wednesday	Grouping and Aggregation
Thursday	Charts and Visualization
Friday	Mini Project – Sales Dashboard Analysis

Weekend Assignment Overview

- Create bar charts and pie charts using datasets.
- Clean missing values from CSV files.
- Analyze sales or student datasets.
- Build a simple dashboard visualization.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 9 – Machine Learning Basics

Week Overview

Students will understand how Machine Learning works and how models are trained using data.

Day	Topics Covered
Monday	Introduction to Machine Learning
Tuesday	Types of Machine Learning
Wednesday	Machine Learning Workflow
Thursday	Introduction to Scikit-learn
Friday	Mini Project – Student Marks Prediction

Weekend Assignment Overview

- Train a simple prediction model.
- Understand datasets and predictions.
- Practice ML terminology and workflow.
- Experiment with simple ML examples.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 10 – ML Models

Week Overview

Students will build beginner-level Machine Learning models and evaluate their performance.

Day	Topics Covered
Monday	Linear Regression
Tuesday	Classification Models
Wednesday	Decision Trees
Thursday	Model Evaluation
Friday	Mini Project – Spam Message Classifier

Weekend Assignment Overview

- Build a basic regression model.
- Train and test classification models.
- Improve prediction accuracy.
- Experiment with different ML datasets.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 11 – Final Project Development

Week Overview

Students will work on their final Machine Learning project with mentor guidance.

Day	Topics Covered
Monday	Project Planning
Tuesday	Dataset Collection
Wednesday	Model Building
Thursday	Testing and Improvements
Friday	Project Completion

Weekend Assignment Overview

- Complete final project development.
- Prepare project screenshots and demo.
- Create GitHub repository for project.
- Document project workflow.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.



Week 12 – Career & Portfolio Development

Week Overview

Students will prepare professional portfolios, resumes, LinkedIn profiles, and present their final projects.

Day	Topics Covered
Monday	Resume Building
Tuesday	LinkedIn Optimization
Wednesday	GitHub Portfolio
Thursday	Mock Interviews & Presentations
Friday	Final Project Demo & Certification

Weekend Assignment Overview

- Update resume with projects and skills.
- Optimize LinkedIn profile professionally.
- Upload final projects to GitHub.
- Prepare final project presentation.

Learning Outcome: By the end of this week students will complete hands-on practical exercises and build confidence through mini projects and guided assignments.

Additional Program Benefits

- One Complete Machine Learning Project
- Certificate of Completion
- Resume Optimization Sessions
- LinkedIn Enhancement
- GitHub Portfolio Creation
- Mini Projects Every Week
- Mock Interview Preparation
- Industry Exposure & Practical Learning

Teaching Methodology

Every lecture follows a practical and interactive structure:

- 10 Minutes – Revision of Previous Concepts
- 20 Minutes – Concept Explanation
- 20 Minutes – Live Coding / Demonstration
- 10 Minutes – Student Practice & Challenges

Final Note

The AI Launchpad Program is focused on building practical skills, problem-solving ability, coding confidence, and real-world exposure for beginners. By the end of the program, students will have hands-on project experience, foundational AI knowledge, portfolio-ready work, and the confidence to continue their learning journey in software development and Artificial Intelligence.